

OMNISTEP FOOTWEAR GLOBAL OPERATIONS REPORT

Prepared By: Paul Tonwe

Collaborating Group: Team Polaris

Date: June 2026

INTRODUCTION

This report delivers a comprehensive, data-backed operational audit of OmniStep, a premier global footwear brand navigating complex omnichannel execution channels. As global retail dynamics increasingly demand seamless channel convergence, maintaining strict visibility over pricing integrity, regional supply chain velocity, and consumer transactional behavior has become a vital necessity for margin preservation.

Currently, OmniStep handles substantial transaction volumes across six primary international boundaries (the United States, the United Kingdom, Germany, India, the United Arab Emirates, and Pakistan) across five distinct footwear performance lines. However, executive leadership has faced significant visibility gaps regarding real-time performance. Aggressive promotional campaigns have historically been leveraged across channels to capture market share, yet the brand lacked the centralized, structural data framework required to verify whether these steep markdown strategies were generating incremental volume or diluting core margins.

The target stakeholders—specifically the CFO, regional supply chain directors, and marketing managers—require a structural transition away from fragmented spreadsheets toward a centralized enterprise data product. To resolve these core analytical requirements, Team Polaris has engineered a high-fidelity business intelligence platform. By refactoring raw data into an optimized database model, this framework isolates margin erosion, identifies localized distribution bottlenecks, and establishes the strategic transparency required to safeguard global profitability margins.

DATA AND METHODOLOGY

The Necessity of a Relational Star Schema

A flat-file data format introduces structural problems, including text redundancy, increased storage overhead, and high risk of transactional update anomalies. Most critically, running calculations against a single flat

table forces the analytical engine to execute full-column scans across millions of repetitive rows for basic aggregations, causing performance lag.

To achieve enterprise-grade data architecture, the database was normalized into a relational **Star Schema**. Normalization separates transactional events from master descriptive attributes. By splitting out high-cardinality strings into separate tables, memory use is drastically reduced, and cross-filtering across visuals becomes fast and secure

Dimension Isolation and Surrogate Key Generation

Through advanced transformation steps in Power Query, two distinct dimensional categories were isolated:

- **Dim_Product**: Contains strict descriptive properties of the inventory assets, including the core category, target consumer gender, shoe color, and shoe size.
- **Dim_Order_Context**: Captures the environment of the purchase, indexing the destination country, the explicit sales channel (Online vs. Retail Store), and the customer affluence cohort.

To decouple the data warehouse from changing source-system operational keys, specialized integer-based **Surrogate Keys** were generated for both tables using sequential indexing to serve as stable Primary Keys (Product_SKU and Context_ID). These lightweight integer columns were then matched back to the master transactional table via relational joins, embedding them as Foreign Keys inside **Fact_Sales**. Once the relational mapping was verified, the bulky descriptive strings were dropped from the fact table, optimizing database compression.

Scalable Calendar Dimension Design

Relying on built-in automatic date features is an anti-pattern that bloats model size and limits analytical flexibility. To enable time-series analysis, a dedicated **Dim_Calendar** dimension table was constructed using dynamic variable boundaries running continuously from January 1, 2018, to December 31, 2026.

This independent dimension houses pre-calculated columns for Year, Quarter, Month Number, Month Name, and Year-Month string stamps. To prevent standard chronological layout sorting errors on the executive

visuals, the text-based Month Name column was structurally bound to the numeric Month Number column using the **Sort by Column** engine, guaranteeing an intuitive January-to-December layout.

Diagnostic DAX Measure Rationale

To support deep business logic beyond simple sums, an 11-measure analytical engine was deployed. The core business logic equations are written as follows:

- **Gross Sales:** Calculates total theoretical revenue before markdown erosion.
- **Net Revenue:** Calculates actual cash performance realized by the business.
- **Effective Discount Rate (Discount Impact):** Measures the proportion of gross revenue sacrificed to promotions, highlighting pricing vulnerabilities.
- **Revenue Retention Rate:** The inverse of discount impact. It acts as an operational health check, measuring how many cents of every gross dollar are successfully kept as net cash.
- **Discount Elasticity Index:** Measures demand responsiveness relative to discount value, evaluating whether a promotion successfully drives volume or simply erodes margin.

KEY FINDINGS

A full audit of OmniStep's transactional data reveals critical performance variances and strategic focus areas across product lines, regional markets, and customer demographics.

=====

=====

GLOBAL SALES PERFORMANCE PROFILE

=====

=====

Total Net Revenue: \$9,083,395.80

Total Units Sold: 75,021

Total Orders Volume: 30,005
13.33%

Average Discount Rate:

Global CSAT Score: 4.0 / 5.0
86.66%

Global Revenue Retention:

1. Profitability Leakage (The Markdown Hemorrhage)

The data shows a distinct margin leak driven by aggressive promotional markdowns. Across all five core shoe lines, baseline performance looks consistent, maintaining a steady Revenue Retention Rate between **86.49% and 86.74%**. However, looking at performance broken down by specific discount tiers reveals a severe discounting anomaly.

Footwear Category & Promo Tier	Total Units Sold	Net Revenue Yield
Basketball (0% Promotion)	2,495	\$342,063.00
Basketball (30% High Promotion)	2,529	\$243,784.10
Running (20% Medium Promotion)	2,640	\$289,120.00
Running (30% High Promotion)	2,471	\$240,829.40

This analysis uncovers two major operational concerns:

- **Severe Margin Cannibalization:** In the *Basketball* category, jumping from a standard full-price model (0% discount) to an aggressive 30% markdown tier only moved 34 additional physical units, while net revenue dropped by **\$98,278.90**.
- **Negative Volume Elasticity:** In the *Running* category, the trend line reverses entirely. Pushing discounts from 20% up to an aggressive 30% tier failed to drive more volume; instead, physical units dropped from 2,640 down to 2,471, leading to an immediate \$48,290.60 loss in net value.

This performance pattern proves that the brand is experiencing a margin leak from high discounts. The *Discount Elasticity Index* drops heavily at the 30% threshold. Furthermore, customer satisfaction (CSAT) stays completely flat at a 4.0 out of 5.0 rating across all transactional records. This indicates that steep discounts do not improve customer sentiment or brand loyalty, serving only to dilute profitability.

2. Customer Demographic Topography

Analyzing consumer spend behavior shows that OmniStep has a remarkably balanced, uniform customer base across all affluence brackets.

Customer Income Level	Total Net Revenue	Total Volume % Share
High Income Cohort	\$3,038,798.30	33.47%
Medium Income Cohort	\$3,003,875.20	33.09%
Low Income Cohort	\$3,040,722.30	33.44%

Net revenue spend is split almost perfectly into thirds, with each affluence bracket contributing roughly \$3M. This balanced distribution carries over to sales channels and payment preferences. Across High, Medium, and Low income cohorts, purchases are evenly split between Online and Retail channels, and each of the four payment methods (Card, Cash, Wallet, Bank Transfer) consistently captures roughly 24% to 25% of total orders. This consistency proves that the brand has strong, broad market appeal,

meaning it does not need to rely on low-price markdowns to attract specific demographics.

3. Geographic & Operational Gaps

While global performance looks healthy, regional execution varies significantly across territories and inventory profiles:

- **Regional Performance Gaps:** The **UAE** serves as OmniStep's top geographic performer, bringing in **\$1,547,084.20** in net value and moving 12,823 units. Conversely, **Pakistan** represents the main regional bottleneck, underperforming the UAE by over **\$80,000** in net revenue and lagging by 760 units.
- **Size Velocity Inconsistencies:** Product size demand is highly uneven. **Size 7** stands out as the primary inventory velocity engine globally, generating **\$1,530,442.40** in net value (12,669 units). In contrast, **Size 8** represents the lowest-velocity asset in the catalog, yielding only **\$1,500,285.35** (12,335 units).

CONCLUSIONS AND RECOMMENDATIONS

Strategic Conclusions

1. **Promotional Inefficiency:** OmniStep's current markdown framework is structurally inefficient. Consumer demand is highly inelastic past the 15% threshold. Pushing markdowns to 30% erodes profit margins without clearing inventory any faster.
2. **Resilient Omni-Channel Demand:** The customer base is exceptionally stable and balanced. The near-identical spending patterns across all income brackets demonstrate a healthy product-market fit that does not require heavy promotional discounting to sustain volume.
3. **Supply Chain Allocation Gaps:** There is a mismatch between regional inventory allocation and localized sizing velocity, resulting in excess capital tied up in lower-velocity products like Size 8 shoes.

Actionable Recommendations

1. Establish Strict Promotional Governance and "Sweet Spot" Caps

- **The Strategy:** Eliminate automated 30% discount options across all online platforms and physical store locations. Transaction data indicates that the optimal promotional "sweet spot" sits between 5% and 15%, where unit volume remains steady while preserving a healthy 86%+ Revenue Retention Rate.
- **The Action:** Enforce a strict 15% promotional cap on standard marketing campaigns to protect margins and prevent price cannibalization.

2. Deploy Dynamic, Time-Gated Markdown Schedules

- **The Strategy:** Transition away from continuous, open-ended promotional campaigns toward a structured, time-gated cadence. Multi-year sales data shows consistent, natural demand surges during specific months (with May, July, and October peaking near \$780K in net value).
- **The Action:** Restrict markdowns during historically high-volume months to capture full-price margins. Limit promotional runs strictly to low-volume windows (such as February shoulder periods) to stimulate artificial demand blocks and clear sitting warehouse inventory.

3. Optimize Supply Chain Allocation and Size Buffers

- **The Strategy:** Adjust inventory distribution and manufacturing runs to align with localized size velocity and regional demand.
- **The Action:** Increase production allocations for high-velocity Size 7 profiles and route them directly to top-performing hubs in the UAE and the UK. Concurrently, reduce inventory buffers for low-velocity Size 8 profiles by 15% to lower holding costs, and implement targeted promotions in Pakistan to close the \$80K regional revenue gap.

4. Expand Data Pipelines to Track Marketing and Advertising Channels

- **The Strategy:** To connect financial leakage directly to customer acquisition costs (CAC), the data warehouse must look beyond transactional sales data.
- **The Action:** Expand the data pipeline by integrating digital marketing sources (such as Google Ads, Meta Campaign Manager, and email click-through records) into the Dim_Order_Context table. Tracking ad

spend alongside explicit transaction IDs will enable Team Polaris to compute accurate Return on Ad Spend (ROAS) and Customer Lifetime Value (CLV) metrics. This addition will provide stakeholders with the visibility needed to evaluate whether aggressive discounting is driven by high customer acquisition costs or inefficient marketing spend